

Shortfall of Domestic Resources to Eradicate Extreme Poverty by 2030

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March 27, 2025

Abstract

In 2015, the Sustainable Development Goals set the eradication of extreme poverty by 2030 as a universally agreed objective. This paper analyses the prospects for achieving this goal country by country. Without a reduction in inequality, even with a very optimistic annual growth rate of 7% between 2022 and 2030, 3% of humans would still be living in extreme poverty in 2030. National capacity to eradicate poverty is then measured using the concepts of *antipoverty cap* or *antipoverty tax* required to finance poverty eradication, and *income floor* (financed by a given income tax). With credible annual growth of 3%, even capping incomes at \$7 a day (to finance an increase in incomes for the poorest individuals) cannot eradicate extreme poverty in 5 low-income countries. In other words, neither growth alone nor growth combined with radical domestic redistribution could eradicate extreme poverty by 2030. By contrast, a transfer of just 0.15% of global income could achieve this goal. This could be financed by a 2% tax on individual net wealth above \$1 billion.

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1 Introduction

The very first Sustainable Development Goal (SDG) reads: “By 2030, eradicate extreme poverty for all people everywhere, currently measured as people living on less than \$2.15 a day”. As we have passed the halfway point since the adoption of the SDGs in 2015, it is time to assess progress towards this universally accepted goal.

In this paper, I assess whether growth and domestic redistribution are sufficient to eradicate extreme poverty by 2030. I first study the extent of poverty in different growth scenarios. Then, I calculate the magnitude of domestic redistribution required in each country to eradicate poverty in 2030. I mobilize different indicators. For two idealized tax policies, I estimate the parameter that would raise enough revenue to eradicate poverty. In the “antipoverty cap”, I fix the tax rate at its maximum value of 100% so that the tax effectively becomes a cap on top incomes, and I find the required cap (or taxation threshold). All income above the cap is assumed to be redistributed to the lowest incomes in order to eradicate poverty. In the “antipoverty tax”, I fix the taxation threshold and find the tax rate needed to raise the revenue required to eradicate poverty. As a last indicator, the “income floor”, I fix both the taxation threshold and the tax rate, and I compute the income floor that the tax could finance (by redistributing tax revenue to the lowest incomes). In the lowest income countries, extreme poverty is estimated to

persist even after strong growth and radical redistribution. This has implications for the international community, as international solidarity appears necessary to achieve the first SDG. I complete the analysis by exemplifying international transfers that would eradicate poverty by 2030.

To assess the prospects of achieving the first SDG, I explore in turn the effects of balanced growth alone, idealized national redistribution policies, and idealized global redistribution. The thought experiments are not meant to be exhaustive or representative pathways to achieving the first SDG. They are presented to demonstrate that relying on growth and domestic policies alone would fall short of eradicating poverty by 2030. This impossibility result is robust to a wide range of growth and redistribution scenarios, including extremely optimistic ones. In contrast, global solidarity has the potential to dramatically accelerate poverty eradication.

Literature The idea to measure the domestic capacity to eradicate poverty with an antipoverty cap originates in [Medeiros \(2006\)](#) (who calls it the “affluence line”). In turn, the idea to measure it with an antipoverty tax dates back to [Ravallion \(2010\)](#) and [Ceriani and Verme \(2014\)](#) (the latter dub it “income lever”). [Ravallion \(2010\)](#) found that even with a 100% tax above the U.S. poverty line, 29 countries could not eradicate extreme poverty, and 37 countries could not eradicate “severe poverty” defined with a higher poverty line (which corresponds to \$3.65/day in 2017 PPP \$). [Bolch, Ceriani, and López-Calva \(2022\)](#) — hereafter “BCL” — update the computations with more recent data and find that 62 countries do not have sufficient resources to eradicate severe poverty.

The present paper employs a similar methodology to assess which countries have sufficient domestic resources to achieve the first SDG. There are three reasons why BCL cannot be used for that purpose. First, the most recent data was not available to BCL (their most recent survey year is 2012 with most years in 2009–2010, compared to 2018–2021 in the present paper). Second, BCL study the data as it stands rather than adjusting it for growth and using it to infer the income distributions in 2030. Third, they focus on a poverty line higher than the one officially used in the first SDG.

In contrast to BCL, I find that 34 countries lack sufficient resources eradicate severe poverty in 2030 in a scenario with 3% growth. Compared to BCL, this less pessimistic finding is largely due to growth up to 2030; though the revision of inequality data also plays a role (see Section 3.3).

Consistently with [Ravallion \(2010\)](#), [Hoy and Sumner \(2016\)](#) find that 52% of global extreme poverty can be eliminated with a 50% antipoverty tax above \$₂₀₁₁10/day (in 2011 PPP). They also consider the reallocation of public spending and show that this antipoverty tax together with the reallocation of fossil-fuel subsidies and military spending could eliminate 77% of global extreme poverty. Finally, they show that countries with GDP per capita below \$₂₀₁₁2,000 per year do not have the domestic capacity to eradicate extreme poverty (measured as an antipoverty tax below 50%).

[Woodward \(2015\)](#) also shows that with growth alone, and if each country’s growth persists at the same level, it would take more than a century — and a global GDP p.c.

exceeding \$100,000 per year — to end extreme poverty. Ortiz, Behrendt, Acuña-Ulate, and Anh (2018) find that financing a basic income at the poverty line is out of reach in low-income countries as the national poverty line is on average equal to 79% of the GDP p.c. and 8 countries have a GDP p.c. below this line (which is often itself below \$2.15/day).

The paper also relates to estimates of future poverty rates based on growth alone. Using GDP projections from the IMF, Karver, Kenny, and Sumner (2012) project the 2030 extreme poverty rate at 2.8% (and at 4.1% if growth is 1% lower than projected, the error historically observed with IMF projections). Other studies are more pessimistic, at 3–7% (Bicaba, Brixiová, & Ncube 2017; Chandy, Ledlie, & Penciakova 2013), 4.7% (Manuel, Desai, Samman, & Evans 2018) or even 7.4% for post-COVID estimates (Lakner, Mahler, Negre, & Prydz 2022).

The paper is linked to the literature that estimates the global income distribution (Alvaredo et al. 2021; Anand & Segal 2015; Fisher-Post & Gethin 2023; Gradín 2021; Jordá & Niño-Zarazúa 2019; Lahoti, Jayadev, & Reddy 2016; Lakner & Milanovic 2016; Milanovic 2024; Pinkovskiy & Sala-i-Martin 2009). I follow Lakner and Milanovic (2016) by merging income and consumption data without adjustment in the main specification. Anand and Segal (2015) discuss common issues regarding the global income distribution and introduce a method to handle discrepancies between data sources. I follow a similar method as a robustness check, as explained in Section 2.

The paper also connects with the costing and progress assessment of SDGs and in particular poverty eradication (Manuel, Carson, Samman, & Evans 2020; Rozenberg & Fay 2019; Schmidt-Traub 2015; SDSN 2019; UN 2022; UNCTAD 2021; Vorisek & Yu 2020). In 24 countries, a growth rate of 7% would not suffice to eradicate extreme poverty by 2030 (UNCTAD 2021). While the global cost of achieving the SDGs may be as high as \$4 trillion per year (UNCTAD 2023), the financing gap in low- and lower-middle-income countries is estimated at \$400 billion (SDSN 2019) to \$700 billion per year (Kharas & McArthur 2019). With the current trend, the SDGs will not be achieved and only limited progress towards them will have been made, with more than 60 countries failing to eradicate extreme poverty by 2030 (Moyer & Hedden 2020). Manuel et al. (2018) find that low-income countries do not have the resources to afford basic healthcare, education, and social protection; only an increase and a redirection of Official Development Assistance (ODA) can finance these programs.

The main contribution of this paper is to provide an explicit assessment of the prospects for achieving the first SDG under ideal circumstances. While existing studies estimate the poverty reduction that can be achieved through either growth or domestic redistribution, I study the combination of both. I also introduce the notion of *income floor* as a measure of the credible potential for redistribution.

The rest of the paper is structured as followed: Section 2 describes the data used in the analysis. Section 3 presents the results: Section 3.1 assesses the effects of growth on poverty; Section 3.2 presents the method to measure domestic capacity to eradicate poverty; Sections 3.3, 3.4, and 3.5 analyze respectively the antipoverty cap, the antipoverty tax, and the income floor; Section 3.6 indicates how global redistribution could

eradicate poverty. Section 4 concludes.

Finally, I provide open source code to work with global inequality data: github.com/bixiou/domestic-poverty-eradication. This code offers ready-to-use functions to compute all sorts of inequality indicators (Gini, top 10% share, antipoverty tax rate, etc.) and plot world maps of these indicators.

2 Data

The percentiles of each country’s post-tax income (or consumption) are estimated by the Poverty and Inequality Platform (PIP) of the World Bank (ex-PovcalNet). PIP aggregates the most recent household surveys (59% of the 167 countries were surveyed between 2018 and 2021). This data is based on purchasing power parity (PPP) and given in constant 2017 \$. PIP data is at the individual level, assuming an equal split of income (or consumption) among all household members, including children.

In low-income countries (those of greatest interest to us), PIP provides data on per capita *consumption* (rather than income). Thereby, the data does not capture services procured by the government. Another potential concern with household surveys is that the aggregate (national) consumption they imply is generally lower than the one estimated in national accounts (Deaton 2005; Hlasny & Verme 2022; Prydz, Jolliffe, & Serajuddin 2022). This discrepancy comes from measurement errors on both sides: on the one hand, household surveys suffer from underreporting of top incomes and large expenditures; on the other hand, national accounts do not properly account for informal work and tend to inflate agricultural output (Angrist, Goldberg, & Jolliffe 2021). Furthermore, authoritarian countries have been shown to produce inflated GDP statistics, except for countries below the GDP threshold of eligibility for preferential loans by the World Bank (Martínez 2022). While Household Final Consumption Expenditures (HFCE) from national accounts is 42% greater than the aggregate consumption from household surveys, the “discrepancy ratio” is largest for middle-income countries and is only 11% for low-income countries. Because household surveys are best suited to estimate consumption by the poorest, I use unadjusted PIP data as a baseline.

As a robustness check, I also re-derive the main results after adjusting aggregate consumption by the discrepancy ratio (computed using World Bank data). In line with Lakner and Milanovic (2013) and Anand and Segal (2015), I impute the extra consumption to the top percentile. I do not perform the rescaling on the 15% of countries (like Burundi or the D.R.C.) with HFCE lower than its aggregate consumption from PIP, and I assume a discrepancy ratio of +11% for the 20% of countries lacking data on HFCE.

As is common in this literature (Bicaba et al. 2017; Hellebrandt & Mauro 2015; Karver et al. 2012), my baseline assumes “balanced growth”, meaning that each percentile grows at the same rate between the country’s survey year and 2030. This assumption allows separating the effects of growth alone from those of a change in the income distribution. I rescale incomes by the observed growth of GDP p.c. (in PPP) up to 2022 (using World Bank data) and by different methods for the 2022–2030 period. These methods include:

extending the 2014–2019 country growth trend (which excludes COVID years); extending the trend for growing countries and assuming no growth when GDP p.c. has contracted between 2014 and 2019; assuming a constant growth (of either 0%, 3%, 4.5%, 6%, or 7%); using IMF forecasts (IMF (2023) extended up to 2030 by replicating the 2026–2028 forecasted growth in 2028–2030); projecting future growth using an autoregressive quadratic model that predicts the 2011–2019 growth based on the 1991–2011 growth (then applied to 2022–2030 using the 2002–2022 growth). Besides, I deviate from this two-step procedure to assess the original SDGs, by assuming a constant growth of 7% starting in 2015. In Supplementary Material, I also analyze a path of unbalanced growth that extends recent trends in national inequality to 2030.

3 Results

3.1 The effect of balanced growth

To estimate global poverty rates, the World Bank scales up the percentiles measured in household surveys by the country’s GDP growth between the survey year and the year of interest.¹ I project global poverty rates and poverty gaps in 2030 using the same assumption of balanced growth (i.e., constant inequality), for a range of growth scenarios (Table 1). I also show that the balanced growth assumption has little effect on poverty estimates compared to an alternative assumption of unbalanced growth in which national inequality evolves as in the recent past (see Supplementary Material for details). This is because inequality has been quite stable over the last years of data, with two thirds of percentile shares growing or contracting by less than 1% per year.

Table 1: Global poverty rates and poverty gaps in 2030 under different growth scenarios. Poverty rates are expressed in % of world population and poverty gaps in % of world GDP. Poverty lines are in 2017 PPP \$/day.

Growth scenario (Poverty line in \$/day)	Poverty rate (%)				Poverty gap (% of GDP)			
	2.15	3.65	6.85	18.15	2.15	3.65	6.85	18.15
2022 Estimate	7.3	21.1	44.4	72.2	0.26	1.36	7.01	42.96
Trend (2014–2019)	6.2	14.4	34.5	66.2	0.21	0.87	4.29	30.64
Max(Trend, 0)	6.3	14.2	34.3	66.4	0.19	0.81	4.16	30.25
Autoregressive projection	6.2	15.2	36.8	65.5	0.17	0.84	4.64	32.02
3% growth	5.2	15.2	37.5	68.2	0.14	0.75	4.38	31.20
3% unbalanced growth	5.1	15.1	37.5	67.7	0.19	0.83	4.48	31.83
7% growth	2.2	8.5	25.5	59.5	0.05	0.29	1.93	18.07
7% growth since 2016	1.1	3.1	15.3	51.3	0.01	0.08	0.74	10.15

My estimates of 2022 global poverty rates align with the 2019 estimates from the World Bank: 9% of the world population live with less than 2.15\$/day, 24% below 3.65\$/day,

and 47% below 6.85\$/day. The poverty gap is the cost that separates people below the poverty line from that line. For example, if 10% of the population earns 1.65\$/day and 90% of the population earns more than 2.15\$/day, the extreme poverty gap is $0.1 \cdot (2.15 - 1.65) = 0.05$ \$/day. I estimate the extreme poverty gap at 0.26% of the world GDP in 2022. This is a first approximation of what it would cost to lift everyone out of extreme poverty, defined with the \$2.15/day poverty line.

Assuming that each country will continue to grow at the same rate as it did in the recent past, I estimate that 6% of the world population will live in extreme poverty in 2030. I find very similar estimates using a simple yet realistic model to predict a country's growth (an autoregressive projection based on its growth over the last 20 years). If each country grows by 3% each year, extreme poverty would decline slightly more than in the realistic projections, at 5%. Although steady growth reduces poverty, growth alone cannot achieve the first SDG: If the world grows by 7% each year (the maximum rate observed for a given country over 2010–2019), the extreme poverty rate would still be 2% in 2030. Even if the world had experienced a 7% growth rate starting in 2015 (when the SDGs were adopted), extreme poverty would not have been completely eliminated, at 1% of the world population in 2030. As we cannot rely on growth alone to eliminate poverty, let us add domestic redistribution to the equation.

3.2 Methodology: idealized redistributive policies

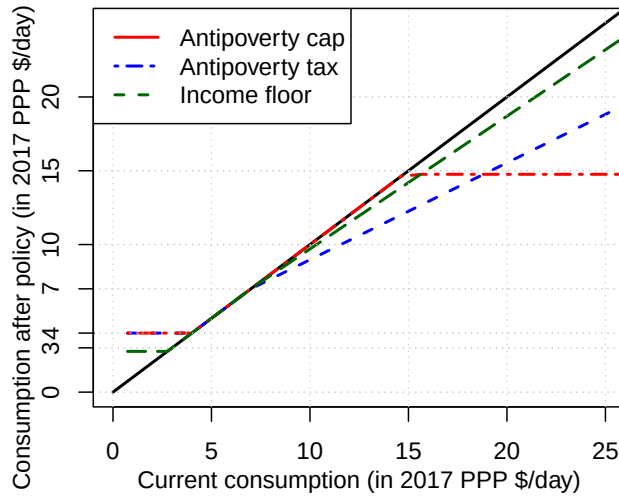
Studying the arithmetics of inequality at the country level, I use the poverty gap to approximate the revenues required to eliminate poverty. More specifically, I consider taxes on top incomes to finance a transfer to the poorest that would lift them at the poverty line. These hypothetical policies would apply on top of existing taxes. I first consider two types of redistributive policies to close the poverty gap: (i) an “antipoverty cap” that would establish a ceiling on top incomes (and tax income at a 100% rate above that threshold); (ii) an “antipoverty tax” that would raise a linear tax above a certain threshold. I also introduce another policy: (iii) the “income floor” that could be funded by a given, realistic tax schedule. The poverty gap can be closed by the latter tax schedule when the income floor exceeds the poverty line.

The idealized policies are depicted in Figure 1 using the example of Kenya: Figure 1a represents the tax and transfers schedule, while Figures 1b, 1c, and 1d show how the policies affect the distribution of income (or rather, consumption). In Figures 1b and 1c, the \$4-poverty gap is the area between the dashed line and the solid line on the left part of the graph. By construction, the area between the two curves on the right part of the graph is also equal to the poverty gap.

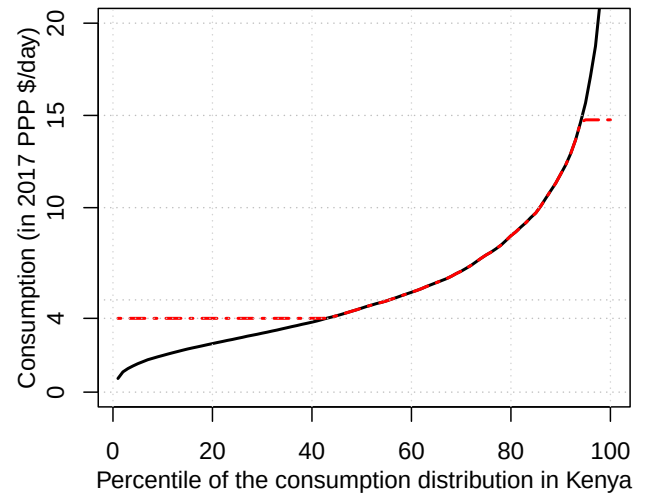
These policies are idealized. The estimate of revenue they generate should be seen as an upper bound of what could be achieved if they were implemented in practice. First, I ignore any costs associated with raising a tax or transferring money, as if the lowest-income countries already had sufficient administrative resources. Second, any tax (and a fortiori a 100% tax) reduces economic activity (real or declared). In this exercise, I abstract

Figure 1: Example of idealized redistributive policies: Kenya in 2030 after 3% growth.

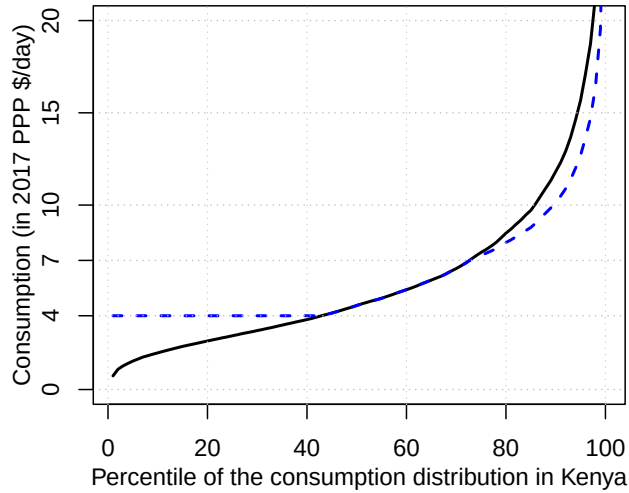
(a) Consumption before/after idealized policies.



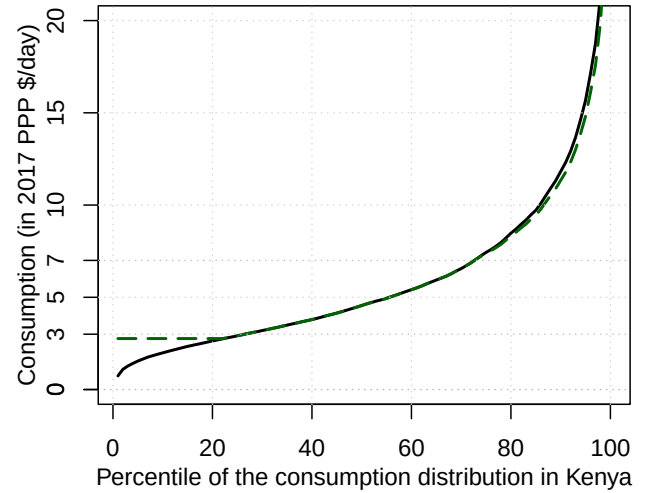
(b) Anti-\$4-poverty cap: \$14.8/day.



(c) Anti-\$4-poverty tax (above \$7/day): 34%.



(d) Income floor (funded by a 10% tax above \$7/day): \$2.8/day.



from tax distortions and assume that the policies would not affect the taxable base.

If it were possible to expropriate top income individuals without reducing their economic activity, capping top incomes to finance an income floor would eliminate poverty at the lowest welfare cost. However, to protect private property and diminish the deterring effect on economic activity, governments would rather tax at a lower rate (than 100%) and on a broader base (starting at a threshold deemed reasonable). Therefore, the antipoverty cap, the antipoverty tax, and the income floor can be thought as rough but revealing approximations of the capacity to mobilize domestic resources. The paper's estimates should not be viewed as causal outcomes of the effects of the idealized policies

simulated.

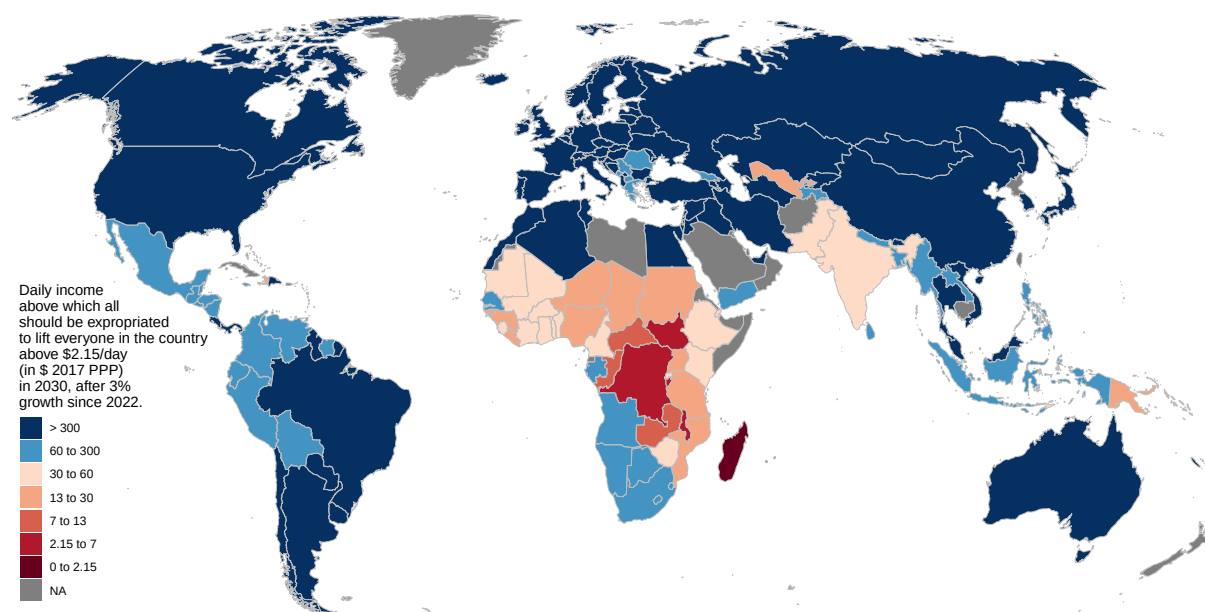
In low-income countries, we measure household consumption rather than income, meaning that we do not capture investment nor government spending. In other words, our idealized policies would leave productive investment and public services unaffected, an appropriate treatment given that these channels already contribute to growth and poverty reduction.

Unless otherwise stated, I use the scenario of balanced growth at a rate of 3%. I choose this rate as a baseline as it is an upper bound of growth rates recently experienced in the lowest-income countries. Indeed, among the 8 countries with an average consumption below \$3/day, growth was on average negative over 2014–2019 (or 2014–2022), and the highest growing country (Central African Republic) grew at a rate of 2.4% per year.

3.3 Antipoverty caps

I estimate the income cap that each country should impose to fill the extreme poverty gap with the expropriated income, i.e. all income above the cap (Figure 2). In some low-income countries, even capping incomes at \$7/day would not suffice to raise revenues equal to the extreme poverty gap, despite a steady growth of 3% per year between 2022 and 2030. In a very optimistic scenario of 7% growth, the anti-*extreme-poverty* cap would be \$14/day in the D.R.C. Also, note that there is no indication that the resources of this country are underestimated, as the aggregate consumption from household surveys is

Figure 2: Income cap eradicating extreme poverty (in \$/day). In this idealized policy, all income above the cap is transferred to the extreme poor and lift them at \$2.15/day, assuming away distortions, and after a yearly growth of 3% over 2022–2030.



greater than HFCE from national accounts for the D.R.C. Besides, the D.R.C. is not the poorest country. In Madagascar, the average consumption would fall short of \$2.15/day in the baseline scenario, at \$2.02/day. This means that even with extreme redistribution, Madagascar does not have the domestic resources needed to eliminate extreme poverty by 2030. To give one last example of the shortfall of resources in the lowest-income countries, the anti-extreme-poverty cap for Burundi in the scenario of 7% growth would need to be as low as 8.60\$/day.

Accounting for country-specific prices of subsistence goods. In most of the paper, I focus on the definition of extreme poverty employed in the first SDG. However, the \$2.15 cut-off has been criticized for inaccurately measuring poverty (Deaton 2010; Deaton & Dupriez 2011; Woodward & Abdallah 2010). First, this poverty line is barely sufficient to satisfy one's caloric requirements and is too low to procure a healthy diet or non-food necessities. Second, the PPP adjustments applied to PIP data before computing the poverty rates are based on prices of an average consumption basket rather than on prices of subsistence goods (Sullivan, Moatsos, & Hickel 2023). Therefore, the cost of a subsistence diet varies across countries. For instance, Moatsos (2016) computes that it is \$1.44 in Malawi vs. \$4.10 in Kenya (in 2011 PPP \$). Building on earlier work by Allen (2017) that addresses these issues, Moatsos (2016) computes a country-specific poverty line. This basic consumption (or BCS) poverty line corresponds to the local price of the cheapest diet that meets caloric and protein requirements, completed with a ration of fat, sugar, and basic non-food requirements (see also Moatsos 2021). This alternative measure, equal to \$4.35/day in median, indicates that poverty is more prevalent than the official poverty line suggests. Despite missing data in many countries (including India and the D.R.C.), 14 countries have an average consumption level below this basic consumption poverty line in 2030 in the 3% growth scenario. These countries (which include e.g. Nigeria) do not have sufficient resources to lift their population above the BCS poverty line even after an extreme domestic redistribution that would equalize all incomes.

Comparing results to previous estimates. BCL found that 62 countries could not eradicate severe poverty (defined as \$₂₀₀₅2/day) with an antipoverty cap at \$₂₀₀₅13/day, while 27 could not even do so with a cap at \$₂₀₀₅2/day. Their findings cannot be exactly reproduced with the revised PIP data, as the switch from 2005 to 2017 PPPs has altered not only the level but also the distribution of incomes (for the same reason, the results of BCL and Ravallion (2010) cannot be compared). When I replicate the computations of BCL (with their survey years but after scaling the original cut-offs into 2017 PPPs by a factor $2.15/1.25 = 1.72$), I find that 53 (resp. 30) countries could not eradicate severe poverty with a cap at \$22.36/day (resp. \$3.44/day). Looking ahead, in our baseline scenario with 3% growth, I find that in 2030, 34 (resp. 6) countries will not be able to eradicate severe poverty with a cap at \$22.36/day (resp. \$3.44/day). In other words, growth significantly reduces poverty and increases the capacity of low-income countries to eradicate poverty.

3.4 Antipoverty taxes

Figure 3 presents the tax rate above \$6.85/day required to generate enough revenues to close the domestic extreme poverty gap, in the baseline scenario of 3% growth. The cut-off of \$6.85/day is defined by the World Bank as the median national poverty line of upper middle-income countries. It corresponds to an “acute” poverty line which can be understood as the consumption level that can sustain a minimally decent life (Hickel 2019; Kikstra, Mastrucci, Min, Riahi, & Rao 2021). In contrast, the extreme poverty line of \$2.15/day corresponds to the consumption per capita below which one is undernourished (Allen 2017).

Consistently with the previous findings, taxing income at a 100% rate above \$6.85/day would not generate enough revenues to eliminate extreme poverty in the five poorest countries. In Nigeria, closing the extreme poverty gap would require taxing the “non-acutely-poor” at a marginal rate of 20%. On average over Sub-Saharan Africa, the anti-*extreme-poverty* tax would be 46%, and 70% in low-income countries (defined by the World Bank as countries with a GNI per capita below \$1,135 per year). Yet, imposing such a large tax burden on any income above just \$6.85/day seems unrealistic.

Figure 4 presents the anti-*extreme-poverty* tax on incomes above \$18.15/day, in a very optimistic scenario of 7% growth. The threshold of \$18.15/day per person corresponds to the U.S. federal poverty line for a family of four and represents a more realistic threshold above which taxes could be increased in the Global South. The anti-*extreme-poverty* tax rates on the “non-poor” in this 7% growth scenario are comparable to the rates on the

Figure 3: Linear tax rate above \$6.85/day eradicating extreme poverty (in %). In this idealized policy, all tax revenue is transferred to the extreme poor and lift them at \$2.15/day, assuming away distortions, and after a yearly growth of 3% over 2022–2030.

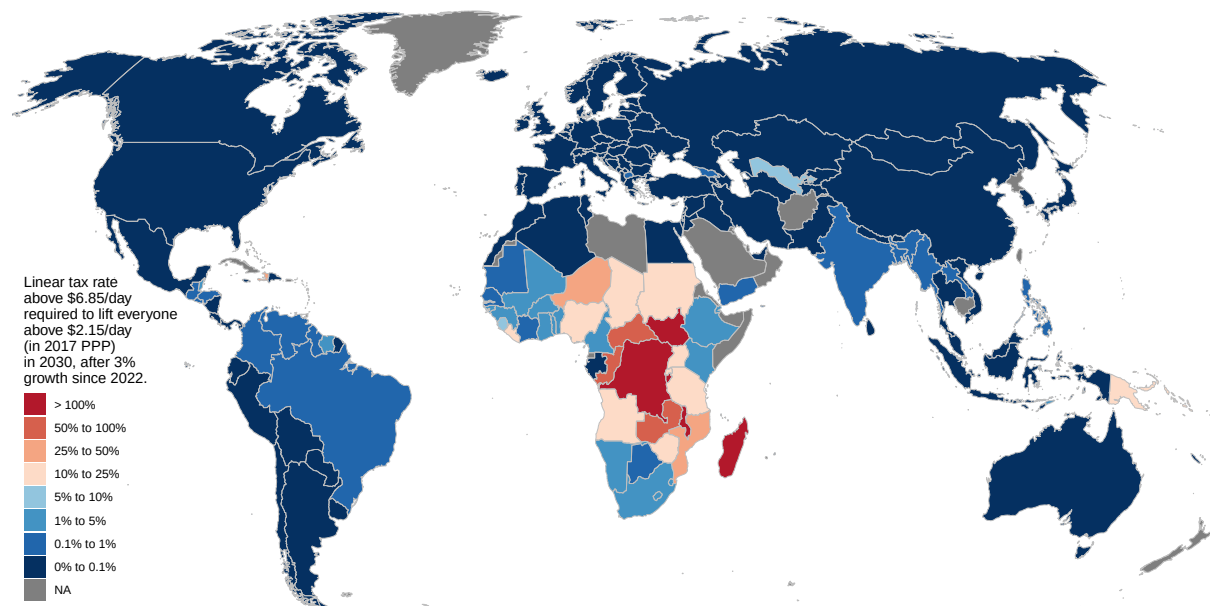
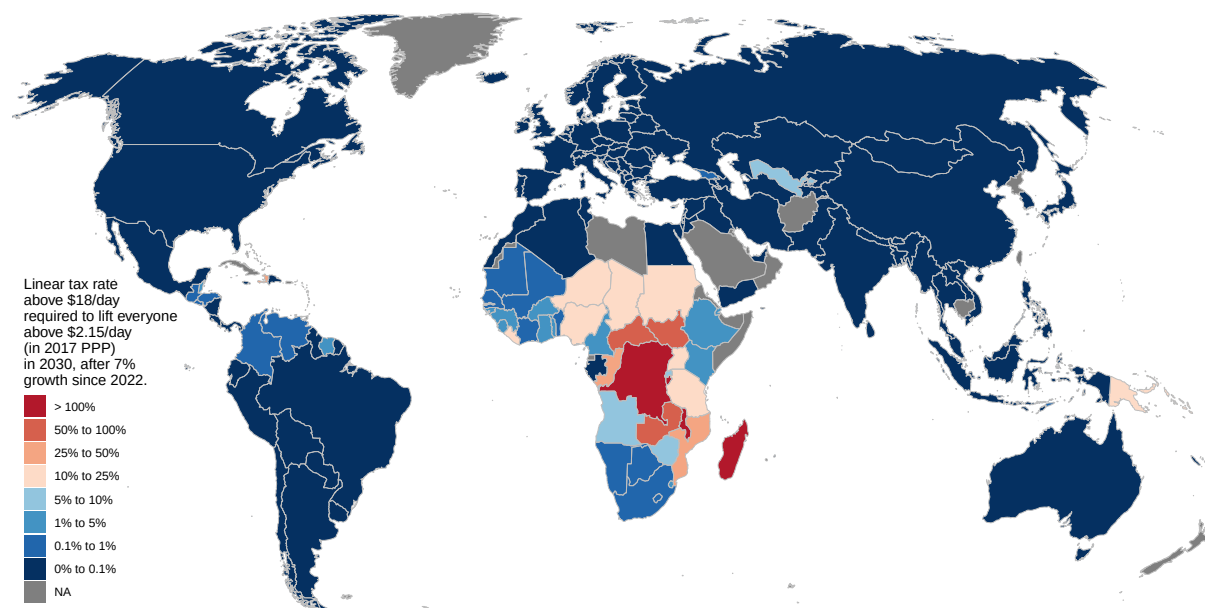


Figure 4: Linear tax rate above \$18.15/day eradicating extreme poverty (in %). In this idealized policy, all tax revenue is transferred to the extreme poor and lift them at \$2.15/day, assuming away distortions, and after a yearly growth of 7% over 2022–2030.



non-acutely-poor in the baseline scenario. In India, the required tax rate would be 10% in the scenario with 7% growth until 2030, 36% with 5.5% growth (the country's 2014–2019 trend), and unachievable (at 156%) with 3% growth. With sustained growth, the contribution required of the Indian non-poor seems large but possible. Therefore, India seems able to eliminate extreme poverty by 2030 with its domestic resources. The same thing cannot be said of Sub-Saharan Africa.

3.5 Income floor as a measure of credible potential for redistribution

A final way of approaching the issue is to set a tax schedule, compute how much revenue it would generate in each country, and estimate the income floor that these revenues could finance (by topping up the incomes of the poorest to the income floor). Note that the value of the income floor depends on the whole income distribution: the top of the distribution determines the revenues that can be generated; and the bottom dictates the cost of raising low incomes up to a given floor. As I have already explored extreme redistributive policies, I analyse here a more reasonable tax schedule. Namely, I consider a 10% marginal tax rate on income above \$6.85/day. Although the tax base I choose may be too wide (affecting people on the verge of acute poverty) and the tax rate too low for top incomes, this simple tax schedule seems to correctly reflect the fiscal capacity of governments. Indeed, this tax schedule is conservatively inspired by [Ravallion \(2010\)](#), who uses a rate of 10% but a higher threshold (corresponding to \$18/day) to assess whether a country has a high or low capacity to eradicate poverty.

Figure 5: Income floor that can be funded with a 10% marginal tax on income above \$6.85/day (in 2017 PPP \$/day). In this idealized policy, all tax revenue is transferred to the poorest and lift them at the income floor, assuming away distortions, and after a yearly growth of 3% over 2022–2030.

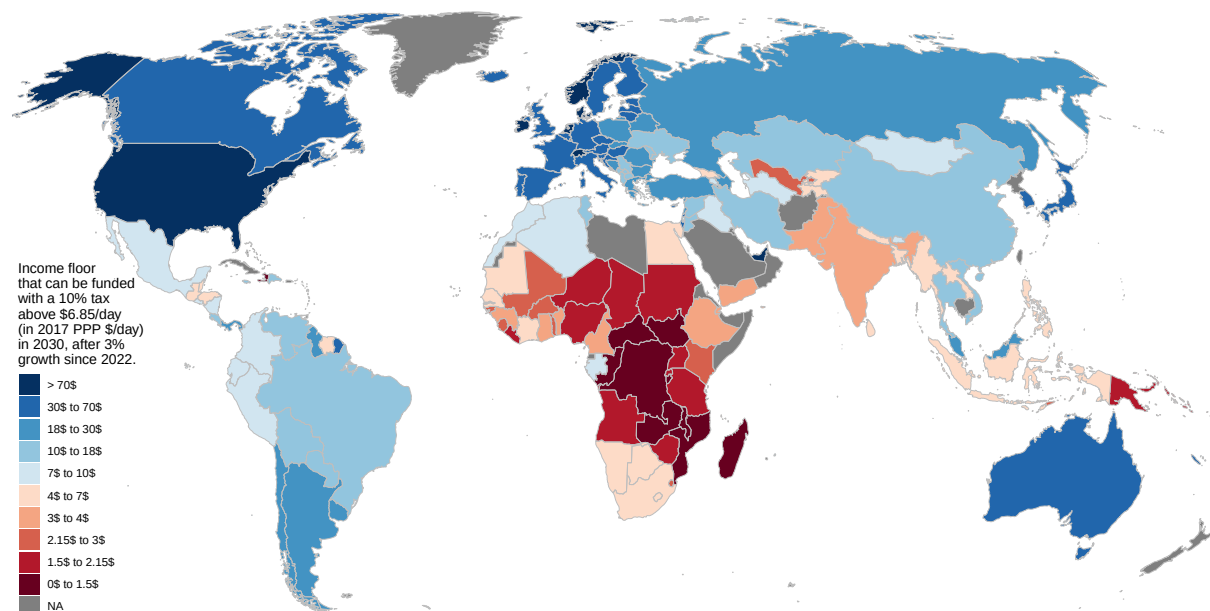


Figure 5 presents the income floor that can be funded in 2030 with our simple tax in a 3% growth scenario. 23 countries are unable to eradicate extreme poverty with this tax. This figure is close to the count of low-income countries, at 27, but only 13 countries fall into both categories. For example, while Ethiopia (a low-income country) can finance an income floor of \$3.08/day, Nigeria (classified as a lower-middle-income country) can only finance a floor of \$1.83/day.

Even in a scenario with 7% growth from 2023 onwards, 10 countries have an income floor below \$2.15 in 2030. Note that the picture does not significantly change when adjusting top incomes so that aggregate consumption matches national accounts: 8 countries are still unable to close the extreme poverty gap despite very optimistic growth in this robustness check. In contrast, if the 7% growth had started in 2016 (as the SDGs were set up), the 10% tax would have been sufficient to eliminate extreme poverty in all countries except in Madagascar, where a tax of 23% would have been required.

Two paths to alleviate poverty. At least two of the SDGs spell out how the elimination of extreme poverty could be funded. First, the target 8.1 aims for “at least 7 per cent gross domestic product growth per annum in the least developed countries”. As we have seen, a sustained high growth since 2016 would have permitted the least developed countries to eliminate extreme poverty through the mobilization of their domestic resources. However, high growth has never materialized in these countries. Second, the target 17.2 calls for “Developed countries to implement fully their official develop-

ment assistance commitments, including the commitment by many developed countries to achieve the target of 0.7 per cent of ODA/GNI to developing countries and 0.15 to 0.20 per cent of ODA/GNI to least developed countries” (LDCs). Foreign aid falls short of both the overall target (at 0.37% of developed countries’ GNI) and the LDCs’ target (at 0.06%). While just four countries are meeting their commitments (Luxembourg, Sweden, Norway, and Germany), the U.S. only allocates 0.23% of its GNI to foreign aid (OECD 2023). The global extreme poverty gap (predicted at 0.17% of global real GDP in 2030) is a bit lower than the shortfall of foreign aid relative to the target (0.2% of global nominal GDP), suggesting that extreme poverty could be significantly reduced if developed countries respected their commitment. However, to meet the broader SDGs and “end poverty in all its forms”, the 0.7% target would not suffice and international solidarity should be significantly strengthened.

3.6 The potential of global redistribution

In this section, we highlight the potential of globally redistributive policies to close the global poverty gap in 2030 in the baseline scenario of 3% growth. The policies described are idealized and should be understood as proxies for any action of international solidarity or system change that would bring about an equivalent transfer of resources.

If applied to the global level, the tax of the previous section would bring the global Gini from .62 down to .51 and finance an income floor at \$8.6/day, thereby closing the \$6.85/day poverty gap. By comparison, applied at the national level, it would only bring the global Gini down to .59 and reduce the poverty gap from 4.5% to 3.7% of global income.

To close the *extreme* poverty gap, a 1.2% marginal tax above 100\$/day (i.e. \$36,500/year) would suffice. Such a tax would result in 0.15% of global income being transferred from the rich to the extreme poor, and almost exclusively transferred from rich countries to poor countries. With contributions of up to 0.4% of a country’s income (in the U.S.), aggregate consumption would increase by more than 10% in 9 countries.

In reality, the global tax rates required to eradicate poverty may well be lower than just indicated, because our calculations used the raw PIP data instead of converting them to nominal terms and rescaling them to national accounts. Once I rescale the data to national accounts (which are more accurate in high-income countries), a mild 0.3% marginal tax above \$100/day suffices to close the extreme poverty gap. Raising that rate to 10% would collect 3.4% of the world income, enough to finance a global income floor of \$7/day and end absolute poverty “in all its forms” (extreme and acute). These rates are expressed on top of the current tax system and apply to post-tax income. For example, the last tax schedule would leave unaffected the 95% of the world’s population whose per capita after-tax income is less than \$36,500/year, and would reduce the after-tax income of those at \$73,000/year by 5%.

Although internationally redistributive taxes have yet to take off, the proposal of a global wealth tax is gaining momentum (Piketty 2022). A 2% tax on individual net wealth

above \$1 billion would raise \$214 billion a year, slightly more than the global extreme poverty gap (Alstadsæter, Godar, Nicolaides, & Zucman 2024). Moreover, a global tax on the wealthiest 1% can raise enough revenues to close the global acute poverty gap and lift everyone above \$6.85/day. For example, the [WID wealth tax simulator](#) shows that a tax consisting of a 4% marginal rate above \$1 million and 10% above \$100 million would raise 4.4% of the global GDP. More generally, a global tax on millionaires designed to be revenue-maximizing in the long-run has the potential to finance the eradication of acute poverty.

While global solidarity has the potential to eradicate poverty, one may wonder whether it is politically and technically feasible. Regarding political acceptability, while government support seems currently unlikely, [Fabre, Douenne, and Mattauch \(2023\)](#) show overwhelming public support for a global tax on millionaires that would finance low-income countries, reaching 69% support in the U.S. and 84% in Europe. As for implementation, government programs offer a proven path to poverty reduction through the deployment of public services, social protection, and infrastructures. Admittedly, our idealized policies entail direct cash transfers to the poorest households. Such transfers are viewed as a promising way to alleviate poverty ([Egger, Haushofer, Miguel, Niehaus, & Walker 2022](#); [Haushofer & Shapiro 2016](#)), but could be challenging to implement. In this regard, progress in payment infrastructures has been encouraging. In particular, the World Bank's *Identification for Development* program finances identification systems in 57 of the world's poorest countries ([World Bank 2017, 2020, 2022](#)), with the aim of "providing legal identity for all" in line with SDG n°16.9. This identification makes it possible to offer numerous services, including cash transfers. In India, the Aadhaar system has provided a unique biometric identifier for 99% of the adult population in just 8 years. Aadhaar is linked to bank accounts and already used to distribute social benefits ([Muralidharan, Niehaus, & Sukhtankar 2023](#)). In less than two weeks, Togo has set up a mobile money transfer for one million people to compensate for the loss of income suffered by informal workers in confined areas during the COVID pandemic ([ipa 2021](#)). Finally, affordable satellite internet and solar panels are making mobile payments accessible in remote areas.

In a nutshell, whereas poverty alleviation cannot be achieved rapidly without international solidarity, it can be financed by reasonable contributions from the global top 1%.

4 Discussion

To paraphrase the [UN \(2022\)](#), "as things stand, the world is not on track to end poverty by 2030." I have shown that the only prospect for low-income countries to eliminate extreme poverty on their own is the combination of strong growth, ambitious social programs and domestic redistribution, and time. With record growth and profound government commitment, China has officially eradicated extreme poverty in 2021. The D.R.C. is poorer today than China was in 1990, so even if the D.R.C. replicates the Chinese miracle, which is not easy given its institutional capacity, the D.R.C. will not be able to eradicate

extreme poverty on its own before 2055. In contrast, international solidarity might be able to end poverty more quickly and at a much lower welfare cost. In this paper, I have illustrated the magnitude of the required transfer of resources with idealized international taxes, but in reality this could take other forms such as a systemic change in the rules or structures of the world economy. To be fair, this paper only presents the orders of magnitude of global inequality. In practice, structural factors that sustain poverty (like wars or corruption) can also hamper the effectiveness of international action.

Finally, a number of limitations prevent the paper from providing definitive estimates of domestic capacities to eradicate poverty. First, while PIP arguably provides the most appropriate data to measure absolute poverty, it does not adequately capture investment, government spending, and top incomes. To better capture in-kind consumption of public services, Fisher-Post and Gethin (2023) offer a promising database, though poverty lines would need to be adjusted accordingly. Second, estimates could be improved when PIP releases a data update, with more recent surveys and prices expressed in 2023 PPP \$. Third, the poverty gap may not be the most accurate measure of the resources needed to eradicate poverty. Instead, it may be more relevant to consider non-economic factors like education and healthcare access, or government programs like social assistance, and assess their cost.

Data and code availability

All data and code of as well as figures of the paper are available on github.com/bixiou/domestic-poverty-eradication. Many more figures (with varying poverty lines, taxation thresholds, growth scenarios, etc.) are available on the repository. Also, any custom figure can be easily produced using this code. The article also comes with a Figshare DOI: [10.6084/m9.figshare.28229477](https://doi.org/10.6084/m9.figshare.28229477).

Notes

¹The World Bank also adjusts growth rates when data captures consumption rather than income, cf. datanalytics.worldbank.org/PIP-Methodology/lineupestimates.html.

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Acknowledgments I am grateful to Michalis Moatsos for his help in using his data. I thank Lidia Cerriani for providing her data. I thank Thomas Goumont and Elise Thai for research assistance.

JEL codes D63, I32, P16.

Keywords Poverty gap, Sustainable Development Goals, Domestic resource mobilization, Redistribution, Fiscal capacity, Extreme poverty.

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A Appendix

A.1 Additional figures

Many more figures (with varying poverty lines, taxation thresholds, growth scenarios, etc.) are available on github.com/bixiou/domestic_poverty_eradication. Also, any custom figure can be easily produced using this code.

Figure A1: Linear tax rate above \$6.85/day eradicating extreme poverty (in %). Data has been rescaled to match HFCE aggregate from national account. In this idealized policy, all tax revenue is transferred to the extreme poor and lift them at \$2.15/day, assuming away distortions, and after a yearly growth of 3% over 2022–2030.

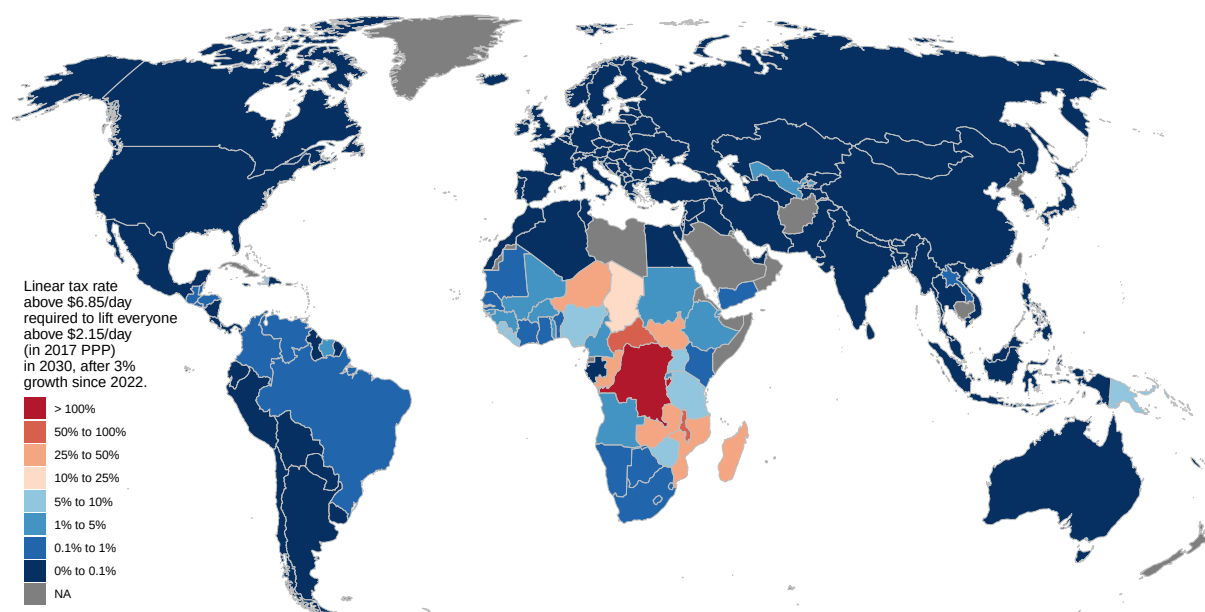


Figure A2: Linear tax rate above \$6.85/day eradicating extreme poverty (in %). In this idealized policy, all tax revenue is transferred to the extreme poor and lift them at \$2.15/day, assuming away distortions, with growth until 2030 predicted at the country level.

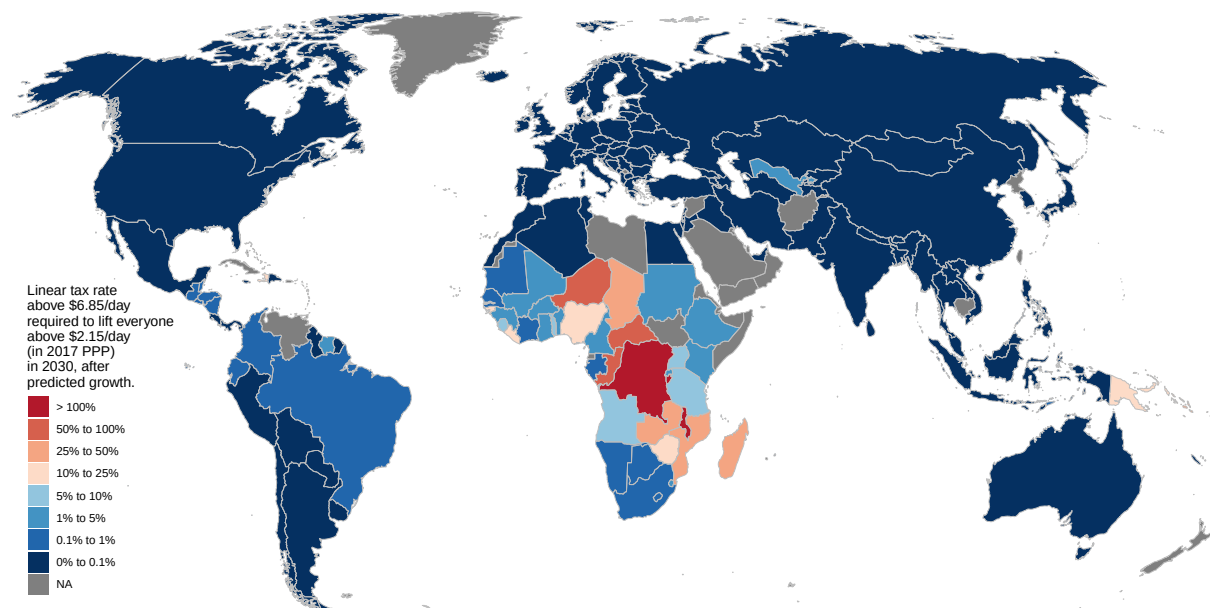


Figure A3: Linear tax rate above \$18.15/day eradicating extreme poverty (in %). In this idealized policy, all tax revenue is transferred to the extreme poor and lift them at \$2.15/day, assuming away distortions, and after a yearly growth of 3% over 2022–2030.

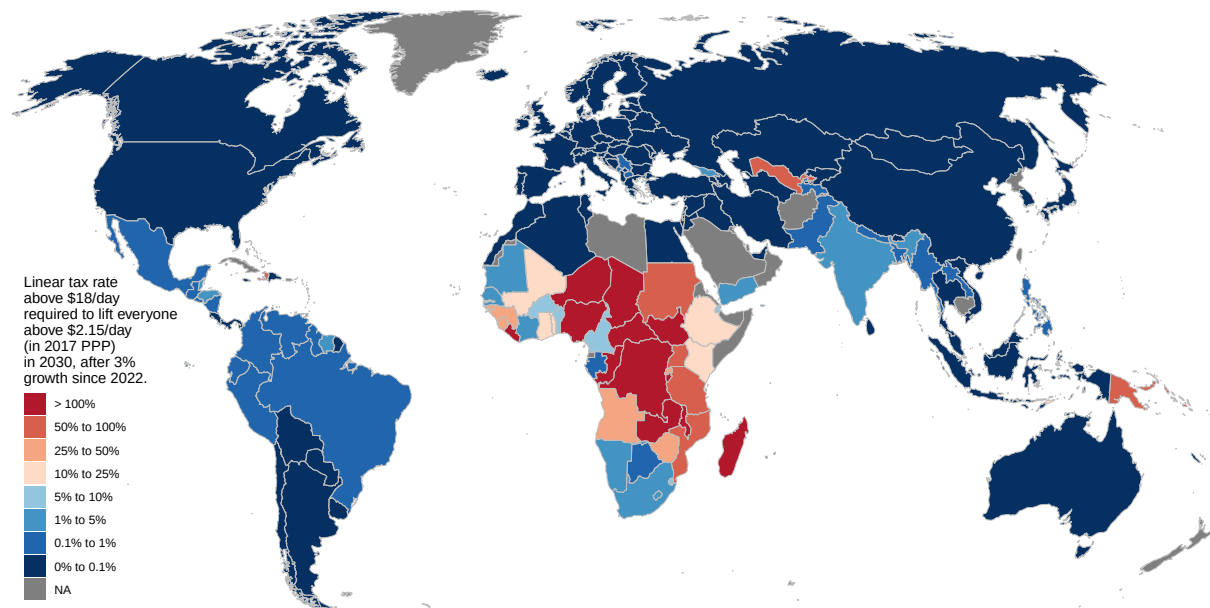


Figure A4: Linear tax rate above \$18.15/day eradicating extreme poverty (in %). Data has been rescaled to match HFCE aggregate from national account. In this idealized policy, all tax revenue is transferred to the extreme poor and lift them at \$2.15/day, assuming away distortions, and after a yearly growth of 7% over 2022–2030.

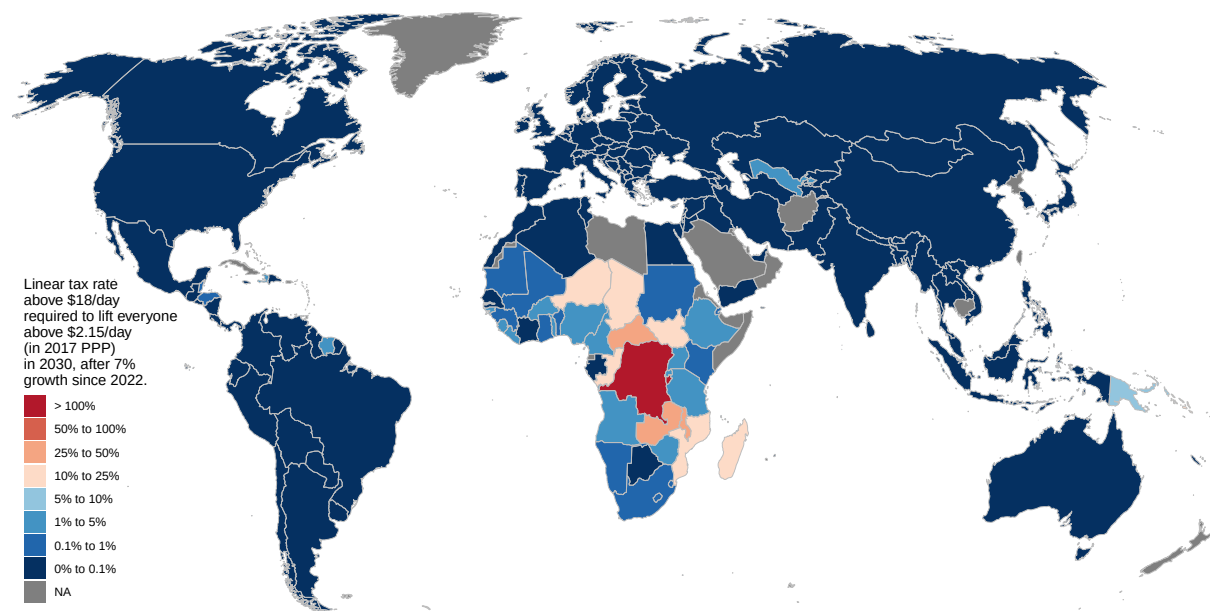


Figure A5: Linear tax rate above \$18.15/day eradicating severe poverty (in %). In this idealized policy, all tax revenue is transferred to the severe poor and lift them at \$3.65/day, assuming away distortions, and after a yearly growth of 3% over 2022–2030.

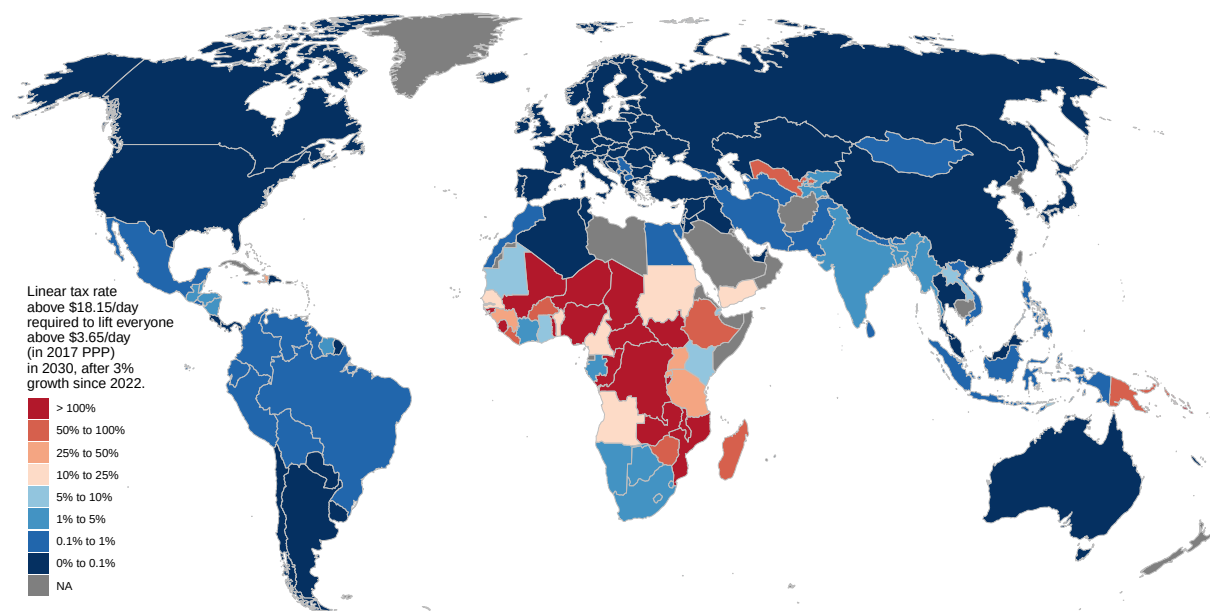


Figure A6: Linear tax rate above \$18.15/day eradicating extreme poverty (in %). In this idealized policy, all tax revenue is transferred to the extreme poor and lift them at \$6.85/day, assuming away distortions, and after a yearly growth of 3% over 2022–2030.

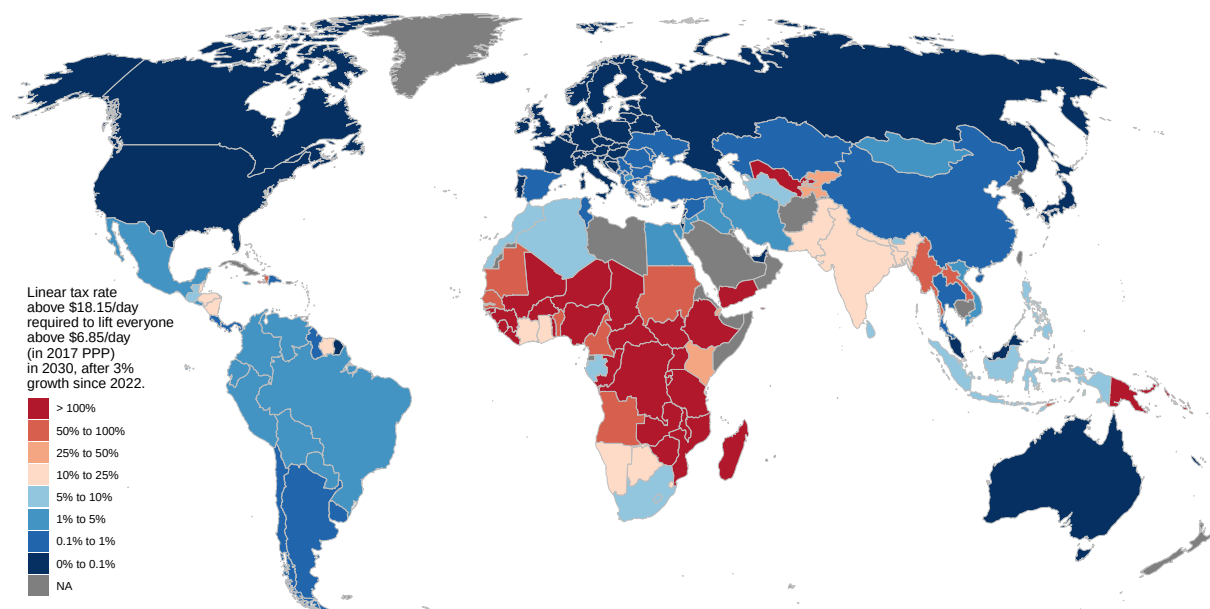


Figure A7: Linear tax rate above \$6.85/day eradicating extreme poverty (in %). In this idealized policy, all tax revenue is transferred to the extreme poor and lift them at \$6.85/day, assuming away distortions, and after a yearly growth of 3% over 2022–2030.

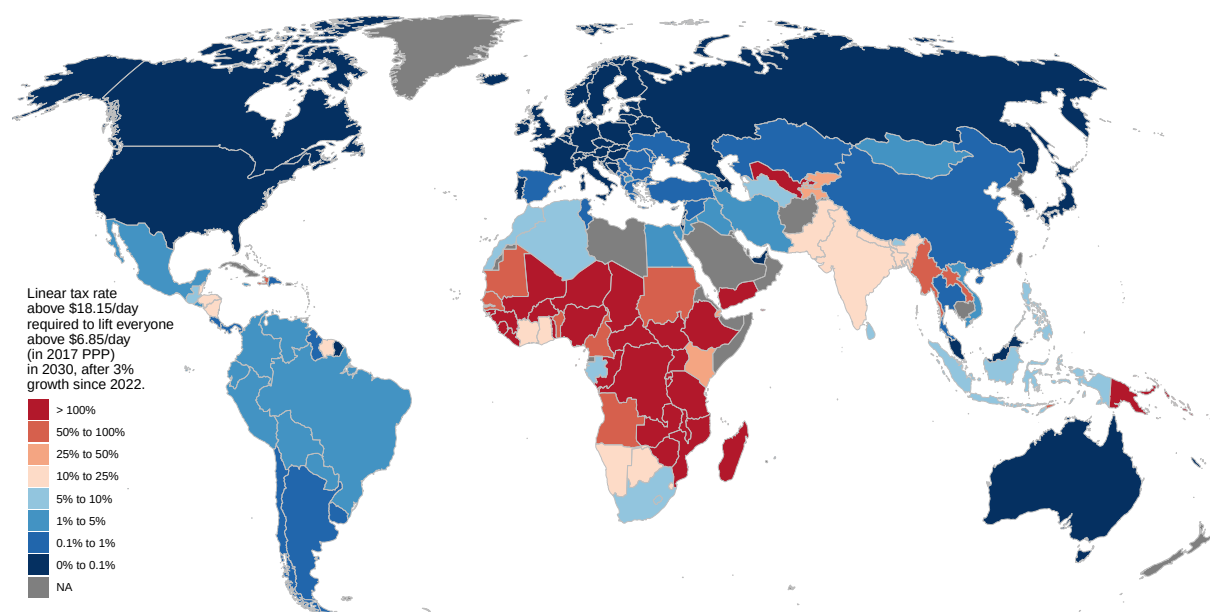


Figure A8: Income floor that can be funded with a 10% marginal tax on income above \$6.85/day (in 2017 PPP \$/day). In this idealized policy, all tax revenue is transferred to the poorest and lift them at the income floor, assuming away distortions, and after a yearly growth of 7% over 2022–2030.

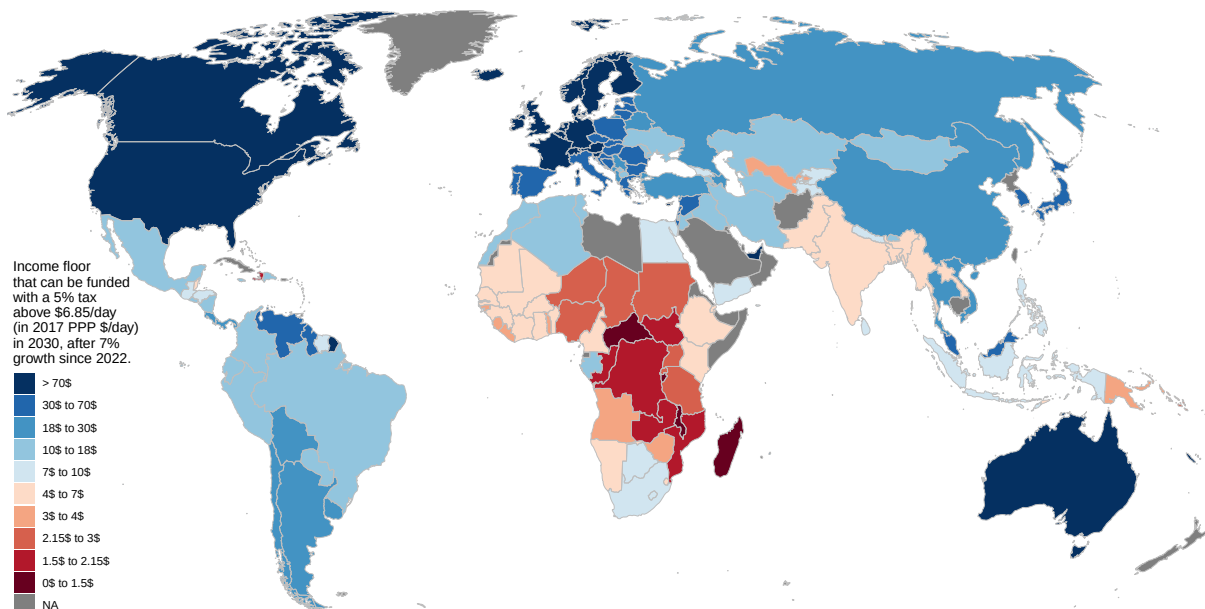
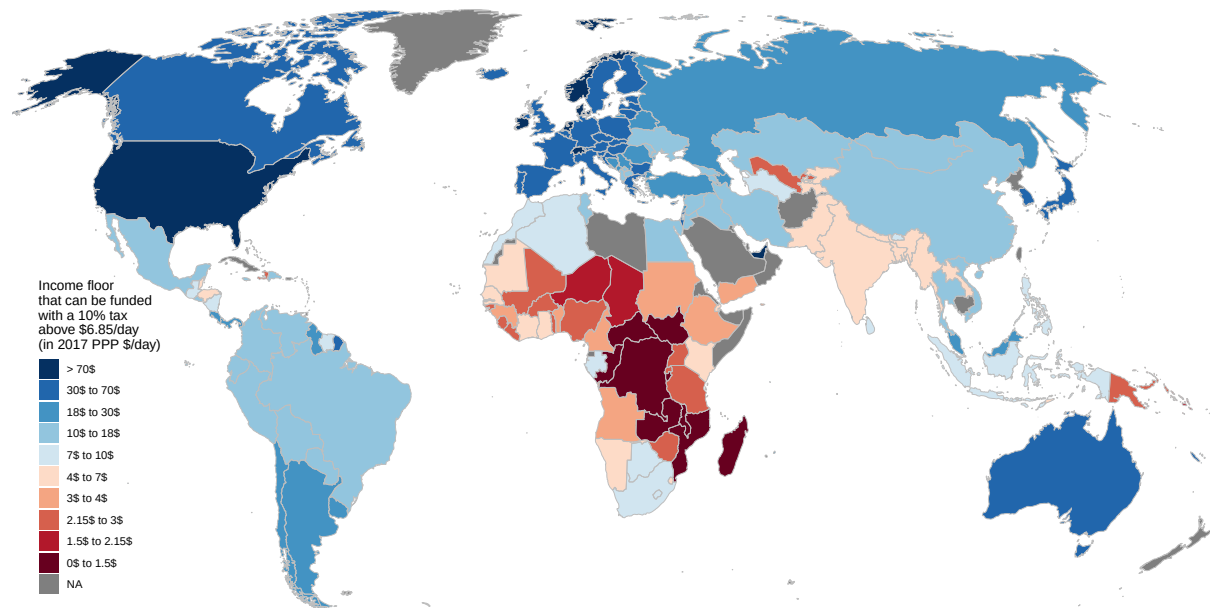


Figure A9: Income floor that can be funded with a 10% marginal tax on income above \$6.85/day (in 2017 PPP \$/day). Data has been rescaled to match HFCE aggregate from national account. In this idealized policy, all tax revenue is transferred to the poorest and lift them at the income floor, assuming away distortions, and after a yearly growth of 3% over 2022–2030.



A.2 Additional tables

Table A1: Mean income in major lower-income countries in various years and growth scenarios, survey years and factor used to rescale incomes to national accounts (in countries with HFCE to survey ratio above 1).

Indicator	Year of Survey	Mean consumption/income (in \$/day) year of survey	2022					BCL survey year	HFCE to survey ratio
			est.	3%	7%	2030 estimate Projection	Trend		
World	2018	16.9	18.3	23.2	31.7	22.5	22.8	2009	1.42
Low-Income Countries	2015	3.6	3.8	5.3	8.5	4.4	4.6	2008	1.11
Sub-Saharan Africa	2016	4.2	4.3	5.5	7.4	5.1	4.8	2009	1.34
Angola	2018	5.5	4.7	6.0	8.1	5.2	3.3	2009	1.71
Bangladesh	2016	4.5	6.1	7.7	10.5	8.0	9.6	2010	1.90
Benin	2018	4.7	5.3	6.7	9.0	6.0	6.1	2003	1.25
Burkina Faso	2018	5.0	5.2	6.6	9.0	6.2	6.5	2009	0.75
Burundi	2013	2.3	2.0	2.5	3.4	2.1	1.6	2006	0.75
Cameroon	2014	5.8	6.1	7.7	10.5	6.8	6.7	2007	1.18
Chad	2018	3.8	3.4	4.4	5.9	3.9	2.6	2003	0.92
D.R. Congo	2012	2.0	2.6	3.2	4.4	3.0	2.8	2005	0.91
Ethiopia	2015	3.7	5.1	6.4	8.7	7.0	8.1	2011	0.91
Ghana	2016	5.1	6.0	7.6	10.3	7.3	7.6	2006	1.96
Guinea	2018	4.4	4.8	6.1	8.3	5.6	6.9	2007	1.12
Haiti	2012	3.9	3.6	4.5	6.2	3.9	3.5	2001	1.73
India	2019	5.3	5.6	7.1	9.7	7.3	8.6	2010	2.19
Ivory Coast	2018	5.7	6.4	8.1	10.9	7.4	8.6	2008	1.67
Kenya	2015	4.2	5.0	6.3	8.6	5.8	6.1	2005	1.99
Madagascar	2012	1.6	1.6	2.0	2.7	1.8	1.7	2010	2.04
Malawi	2019	2.1	2.0	2.5	3.4	2.3	2.2	2010	NA
Mali	2018	5.0	4.9	6.2	8.3	5.4	5.7	2010	0.94
Mozambique	2014	2.9	2.9	3.7	5.0	3.4	3.1	2008	NA
Nepal	2010	5.2	7.7	9.8	13.3	9.4	11.0	2010	1.30
Niger	2018	2.9	3.0	3.9	5.2	3.5	3.5	2008	0.86
Nigeria	2018	3.7	3.6	4.5	6.1	4.1	3.2	2011	NA
Pakistan	2018	5.0	5.3	6.7	9.1	6.2	6.8	2008	2.35
Papua New Guinea	2009	3.5	4.3	5.4	7.3	4.8	4.8	1996	NA
Rwanda	2016	3.1	3.8	4.8	6.6	4.9	5.5	2011	1.45
Senegal	2018	6.0	6.3	8.0	10.8	7.2	8.2	2011	1.11
South Sudan	2016	2.2	2.2	3.3	5.5	NA	2.2	NA	NA
Sudan	2014	4.6	3.4	4.3	5.9	3.6	2.7	2009	1.94
Tanzania	2018	3.3	3.4	4.4	5.9	4.1	4.3	2007	1.43
Uganda	2019	3.5	3.6	4.5	6.1	4.2	4.1	2009	1.33
Uzbekistan	2003	1.6	4.0	5.1	6.9	5.2	5.5	NA	NA
Yemen	2014	4.5	4.5	7.2	13.3	NA	4.5	2005	NA
Zambia	2015	3.0	3.0	3.7	5.1	3.4	2.9	2010	NA
Zimbabwe	2019	4.5	4.4	5.5	7.5	4.7	4.0	NA	1.10

Table A2: Expected poverty and growth in major lower-income countries: trend and projected growth rate, poverty rates and gaps at \$2.15 and \$6.85/day in 2030 after 3% growth since 2022.

Indicator	Growth Trend 2014–2019	Growth Autoregressive Projection	Poverty rate (in %)			Poverty gap (in % of mean income)		
			\$2.15	\$3.65	\$6.85	\$2.15	\$3.65	\$6.85
World	3.3	2.9	5	15	38	0.1	0.8	4.5
Low-Income Countries	1.9	2.3	27	52	80	3.9	15.5	56.7
Sub-Saharan Africa	1.0	2.0	24	50	79	3.2	13.6	52.9
Angola	-4.2	1.3	28	49	75	3.7	13.6	47.9
Bangladesh	5.8	3.4	1	13	58	0.0	1.1	16.0
Benin	1.8	1.7	7	29	69	0.4	4.3	29.3
Burkina Faso	2.7	2.1	15	45	73	1.1	8.1	37.9
Burundi	-2.6	0.9	60	84	96	19.0	63.9	181.6
Cameroon	1.2	1.5	15	35	62	1.2	6.1	26.7
Chad	-3.4	1.7	23	57	86	2.8	16.9	71.5
D.R. Congo	1.2	2.1	45	73	91	11.2	39.5	122.7
Ethiopia	6.0	4.0	7	26	71	0.5	4.1	28.9
Ghana	2.9	2.5	13	30	61	1.3	5.5	24.9
Guinea	4.6	1.9	4	25	69	0.3	3.6	29.3
Haiti	-0.3	1.1	37	59	83	8.2	24.4	76.1
India	5.5	3.4	3	21	68	0.1	2.3	23.3
Ivory Coast	3.9	1.9	3	19	56	0.1	2.0	17.7
Kenya	2.6	1.9	12	37	71	1.0	6.8	35.5
Madagascar	1.1	1.3	72	88	97	35.4	96.0	244.4
Malawi	1.0	1.8	59	84	96	18.1	61.7	177.3
Mali	2.1	1.4	7	34	71	0.4	5.2	34.1
Mozambique	0.9	2.2	52	76	90	12.3	39.2	113.1
Nepal	4.5	2.5	1	5	40	0.0	0.4	7.5
Niger	1.7	1.6	28	67	90	3.8	23.1	90.9
Nigeria	-1.3	1.8	20	50	84	2.4	14.3	64.1
Pakistan	3.2	2.0	1	15	68	0.0	1.3	22.4
Papua New Guinea	1.5	1.6	21	45	76	2.9	12.2	49.3
Rwanda	4.8	3.0	25	57	84	2.9	15.9	64.2
Senegal	3.3	1.7	3	20	58	0.1	2.0	18.5
South Sudan	NA	NA	47	72	91	12.4	40.5	122.8
Sudan	-2.9	0.8	18	54	88	2.1	14.9	70.5
Tanzania	2.8	2.3	26	61	87	3.1	18.7	74.9
Uganda	1.7	2.2	27	59	85	3.6	18.3	71.7
Uzbekistan	4.0	3.3	12	43	82	1.4	9.4	51.5
Yemen	NA	NA	4	23	64	0.2	2.9	23.2
Zambia	0.0	1.9	53	71	87	15.4	40.8	109.6
Zimbabwe	-1.0	0.9	29	55	79	3.3	15.1	55.1

Table A3: Antipoverty caps for major lower-income countries in 2030.

Poverty line (\$/day)	2.15	2.15	2.15	2.15	2.15	2.15	BCS	3.44	3.44
Growth scenario	3%	3%	7%	7%	Projection		3%	3%	BCL
HFCE rescaling		✓		✓		✓			
World	442.6	1785.7	618.7	2442.4	384.3	1741.5	$+\infty$	376.0	33.2
Low-Income Countries	29.7	197.0	160.9	498.9	15.7	99.5	$+\infty$	13.1	2.9
Sub-Saharan Africa	37.7	635.4	52.8	1119.1	31.0	631.5	$+\infty$	16.3	3.8
Angola	63.1	896.5	132.4	1283.4	41.1	755.7	6.0	20.7	19.6
Bangladesh	69.0	1461.6	$+\infty$	$+\infty$	71.4	1506.2	NA	50.1	5.8
Benin	50.1	376.6	79.0	523.0	40.1	333.0	30.5	24.4	2.8
Burkina Faso	49.4	49.4	90.5	90.5	40.9	40.9	24.7	25.1	2.8
Burundi	3.4	3.4	8.6	8.6	2.1	2.1	2.5	2.5	2.0
Cameroon	53.8	326.9	97.0	476.0	40.4	275.0	17.3	29.5	7.2
Chad	16.4	16.4	34.2	34.2	12.4	12.4	10.1	6.9	2.5
D.R. Congo	6.7	6.7	14.7	14.7	5.6	5.6	NA	3.2	1.0
Ethiopia	43.8	43.8	71.7	71.7	51.2	51.2	19.2	22.0	3.4
Ghana	47.8	1501.7	85.1	2067.0	43.8	1440.7	30.1	26.3	3.4
Guinea	21.6	160.4	35.0	225.3	18.2	143.5	17.2	13.2	3.0
Haiti	18.5	593.3	39.5	895.7	12.1	466.8	4.5	7.9	2.7
India	56.1	1750.5	79.1	2378.6	58.3	1804.3	NA	35.3	4.2
Ivory Coast	55.4	1127.2	80.2	1534.2	48.5	1028.7	NA	32.5	5.3
Kenya	42.1	1291.1	75.4	1775.5	33.6	1174.5	14.1	19.3	6.3
Madagascar	2.0	200.7	4.6	422.2	1.8	125.5	2.0	2.0	1.7
Malawi	3.6	6.7	9.1	36.8	2.7	4.3	8.2	2.5	2.3
Mali	33.8	33.8	55.6	55.6	25.3	25.3	27.1	16.9	2.8
Mozambique	21.0	54.5	59.5	153.1	15.6	40.4	28.9	4.5	2.1
Nepal	66.3	660.2	$+\infty$	$+\infty$	63.5	635.2	38.2	56.9	10.8
Niger	16.2	16.2	38.4	38.4	11.3	11.3	8.0	5.0	1.8
Nigeria	14.2	89.7	29.5	159.9	11.1	68.7	5.9	7.0	3.6
Pakistan	55.9	1875.4	$+\infty$	$+\infty$	51.2	1740.1	20.1	36.3	5.4
Papua New Guinea	20.6	108.1	39.2	190.6	16.1	82.0	NA	11.1	3.0
Rwanda	32.0	451.2	74.1	664.8	32.3	452.6	7.2	10.9	2.9
Senegal	72.7	250.6	103.0	344.4	62.9	223.3	60.8	41.7	3.4
South Sudan	6.8	17.0	24.9	120.3	NA	NA	NA	3.3	NA
Sudan	20.0	819.5	49.0	1148.3	11.5	661.9	4.3	6.7	7.5
Tanzania	23.7	378.8	57.8	567.6	19.4	345.9	9.6	7.5	2.7
Uganda	29.9	314.0	77.3	483.3	23.3	281.0	10.4	8.5	3.4
Uzbekistan	22.7	124.3	44.9	193.6	24.1	129.7	36.2	10.8	NA
Yemen	60.2	216.7	$+\infty$	$+\infty$	NA	NA	NA	30.2	12.8
Zambia	11.8	27.8	27.5	80.2	9.3	20.0	3.7	4.6	2.6
Zimbabwe	40.6	133.1	85.3	235.7	26.7	83.7	NA	17.2	NA

Table A4: Antipoverty tax required to eliminate extreme poverty (at \$2.15/day) in major lower-income countries in 2030 (marginal rate in %).

Taxation threshold (\$/day)	6.85	18.15	18.15	18.15	18.15	6.85	6.85	6.85	18.15
Growth scenario	3%	7%	3%	Trend	Projection		7%	3%	7%
HFCE rescaling								✓	✓
World	0.2	0.1	0.3	0.4	0.3	0.2	0.1	0.1	0.1
Low-Income Countries	14.6	5.5	44.5	112.2	125.9	31.0	2.9	9.6	3.5
Sub-Saharan Africa	11.5	8.3	30.1	61.5	42.5	15.5	3.3	5.1	2.5
Angola	11.4	9.1	28.2	177.2	44.3	18.0	3.8	3.6	1.8
Bangladesh	0.1	0.0	0.6	0.0	0.4	0.1	0.0	0.0	0.0
Benin	1.5	0.7	6.8	12.8	13.8	3.0	0.2	0.8	0.2
Burkina Faso	3.2	1.3	9.7	11.1	14.7	4.7	0.5	3.2	1.3
Burundi	285.8	449.4	> 10k	> 10k	> 10k	600.7	69.5	285.8	449.4
Cameroon	3.1	2.0	9.9	19.6	17.9	5.3	0.8	2.1	1.0
Chad	19.3	16.1	130.6	> 10k	246.0	33.9	3.2	19.3	16.1
D.R. Congo	103.1	155.2	743.7	2482.0	1239.1	140.8	26.8	103.1	155.2
Ethiopia	2.2	1.1	10.3	2.0	5.9	1.3	0.3	2.2	1.1
Ghana	3.6	3.5	13.9	14.3	16.6	4.2	1.1	1.0	0.5
Guinea	1.8	2.1	36.0	10.4	85.3	3.3	0.2	1.0	0.4
Haiti	32.7	34.3	97.3	242.0	164.6	52.9	12.5	8.3	4.9
India	0.5	0.1	1.6	0.4	1.2	0.4	0.0	0.1	0.0
Ivory Coast	0.4	0.2	2.0	1.2	3.9	0.8	0.1	0.1	0.0
Kenya	3.8	2.8	15.9	19.5	25.9	5.9	0.8	0.8	0.3
Madagascar	620.6	2019.5	> 10k	> 10k	> 10k	1072.2	179.7	32.3	16.4
Malawi	260.0	354.8	> 10k	> 10k	> 10k	397.8	63.8	101.4	55.3
Mali	1.9	0.9	12.0	21.2	30.9	4.3	0.2	1.9	0.9
Mozambique	45.8	32.0	89.2	151.8	109.9	56.1	16.6	32.6	19.5
Nepal	0.0	0.0	0.1	0.0	0.2	0.0	0.0	0.0	0.0
Niger	28.5	17.6	123.5	222.1	228.3	50.2	4.7	28.5	17.6
Nigeria	19.5	22.7	187.6	> 10k	387.0	33.7	3.0	10.4	4.6
Pakistan	0.0	0.0	0.3	0.1	1.0	0.2	0.0	0.0	0.0
Papua New Guinea	13.2	15.2	75.0	136.5	128.1	20.9	3.5	8.8	6.2
Rwanda	12.9	8.2	41.2	20.8	40.7	12.7	2.8	4.3	1.5
Senegal	0.4	0.2	1.5	1.2	3.1	0.8	0.1	0.3	0.1
South Sudan	101.6	52.3	603.5	> 10k	NA	NA	11.7	53.0	17.6
Sudan	17.8	12.4	73.0	2942.7	197.9	46.4	2.8	2.0	0.6
Tanzania	17.7	10.0	62.2	68.1	86.3	23.7	3.0	5.1	1.5
Uganda	18.3	12.1	52.4	81.4	69.8	24.4	4.1	6.8	2.7
Uzbekistan	8.5	8.1	51.7	31.7	44.2	7.5	1.6	5.1	2.6
Yemen	0.9	0.0	3.8	86.1	NA	NA	0.0	0.6	0.0
Zambia	58.2	60.1	166.2	362.1	216.7	74.3	23.7	41.2	33.3
Zimbabwe	10.5	6.1	27.2	107.9	56.9	20.5	2.6	7.8	3.7

Table A5: Antipoverty tax required to eliminate severe poverty (at \$3.65/day) in major lower-income countries in 2030 (marginal rate in %).

Taxation threshold (\$/day)	6.85	18.15	18.15	18.15	18.15	6.85	6.85	6.85	18.15
Growth scenario	3%	7%	3%	Trend	Projection		7%	3%	7%
HFCE rescaling								✓	✓
World	1.0	0.5	1.4	1.7	1.6	1.1	0.4	0.7	0.3
Low-Income Countries	57.2	25.1	174.5	388.0	464.3	114.4	13.4	37.7	16.1
Sub-Saharan Africa	49.2	42.2	129.2	224.3	172.5	63.1	16.7	22.0	12.5
Angola	41.2	38.0	102.3	510.7	151.6	61.6	15.8	13.0	7.5
Bangladesh	3.8	1.7	19.0	3.5	15.2	3.1	0.4	0.9	0.2
Benin	16.4	12.3	72.2	118.6	126.0	27.1	3.3	8.5	3.5
Burkina Faso	23.4	16.4	70.8	78.6	97.3	30.9	6.6	23.4	16.4
Burundi	963.8	1983.9	> 10k	> 10k	> 10k	1787.8	306.7	963.8	1983.9
Cameroon	16.2	13.8	51.6	91.0	84.4	24.7	5.2	11.0	6.9
Chad	116.1	146.6	785.2	> 10k	1304.4	179.7	29.0	116.1	146.6
D.R. Congo	362.1	659.4	2611.6	8008.2	4168.2	473.6	113.8	362.1	659.4
Ethiopia	18.4	16.4	85.8	24.8	55.7	12.4	4.0	18.4	16.4
Ghana	15.8	16.1	60.3	62.0	71.3	18.2	5.2	4.2	2.1
Guinea	21.5	35.3	441.1	152.0	929.6	36.5	3.3	12.7	6.7
Haiti	97.1	108.3	288.8	681.1	473.9	152.1	39.3	24.8	15.4
India	8.3	3.8	28.6	8.3	23.6	7.0	1.2	1.5	0.3
Ivory Coast	6.3	3.8	28.6	18.1	49.5	10.1	1.1	2.1	0.6
Kenya	25.2	22.5	106.9	127.0	162.2	36.6	6.4	5.4	2.2
Madagascar	1683.2	6551.1	> 10k	> 10k	> 10k	2730.1	582.8	87.7	53.4
Malawi	886.5	1596.0	> 10k	> 10k	> 10k	1260.5	287.0	345.8	248.8
Mali	23.1	19.5	149.6	232.4	312.3	43.2	4.1	23.1	19.5
Mozambique	146.1	122.9	284.4	441.9	337.8	172.4	64.0	104.0	75.0
Nepal	1.0	0.5	4.7	2.1	6.2	1.3	0.1	0.5	0.2
Niger	172.7	171.6	747.5	1165.2	1190.4	261.7	46.1	172.7	171.6
Nigeria	115.2	197.4	1107.3	> 10k	2067.2	179.9	26.0	61.3	40.3
Pakistan	6.5	2.2	35.3	30.8	62.2	11.6	0.4	0.9	0.1
Papua New Guinea	54.8	74.0	312.4	540.0	509.4	83.0	17.0	36.8	30.0
Rwanda	69.4	61.7	222.3	129.5	220.5	68.8	20.9	23.4	11.5
Senegal	6.2	3.2	23.7	20.1	44.0	10.9	1.0	4.7	1.8
South Sudan	333.2	216.6	1978.2	> 10k	NA	NA	48.3	173.9	72.9
Sudan	124.1	120.0	509.8	> 10k	1125.3	263.7	27.4	14.1	5.4
Tanzania	105.4	96.6	370.5	396.7	475.1	130.7	29.3	30.6	14.4
Uganda	91.9	81.8	263.4	372.6	331.2	115.9	28.0	34.3	18.0
Uzbekistan	57.2	62.5	347.6	220.2	300.4	50.7	12.7	34.4	20.0
Yemen	10.3	0.9	45.3	560.7	NA	NA	0.3	7.4	0.6
Zambia	154.0	176.1	440.1	898.2	560.3	192.0	69.3	109.1	97.4
Zimbabwe	47.7	39.4	123.6	375.2	221.7	80.1	16.9	35.5	24.1

Table A6: Antipoverty tax required to eliminate acute poverty (at \$6.85/day) in major lower-income countries in 2030 (marginal rate in %).

Taxation threshold (\$/day)	6.85	18.15	18.15	18.15	18.15	6.85	6.85	6.85	18.15
Growth scenario	3%	7%	3%	Trend	Projection		7%	3%	7%
HFCE rescaling								✓	✓
World	5.8	3.1	8.1	8.1	8.8	6.2	2.4	4.0	2.0
Low-Income Countries	210.0	107.8	640.1	1269.2	1592.3	392.5	57.5	138.3	69.2
Sub-Saharan Africa	190.8	198.4	501.4	753.1	634.2	231.9	78.5	85.5	59.0
Angola	145.1	153.3	360.6	1450.0	506.1	205.7	63.8	45.8	30.2
Bangladesh	58.3	54.6	289.5	91.3	250.1	50.9	13.0	13.6	5.2
Benin	110.9	129.5	489.2	713.7	747.4	160.7	34.8	57.4	37.3
Burkina Faso	109.5	111.8	331.0	358.6	422.7	134.1	45.0	109.5	111.8
Burundi	2738.5	6734.8	> 10k	> 10k	> 10k	4724.1	1041.1	2738.5	6734.8
Cameroon	70.6	75.5	224.7	365.9	342.8	100.4	28.5	47.7	37.7
Chad	491.2	858.6	3323.0	> 10k	5051.1	695.7	170.0	491.2	858.6
D.R. Congo	1125.8	2436.1	8119.6	> 10k	> 10k	1422.3	420.2	1125.8	2436.1
Ethiopia	128.1	147.4	597.2	208.3	415.6	92.5	36.1	128.1	147.4
Ghana	71.4	83.1	272.4	279.2	317.1	80.8	26.9	18.9	11.1
Guinea	175.2	444.1	3599.3	1477.9	6695.5	262.8	41.2	103.7	83.7
Haiti	303.5	368.9	902.7	1963.9	1412.4	453.5	133.9	77.4	52.5
India	85.5	71.2	294.9	124.4	258.1	75.9	23.1	15.7	6.6
Ivory Coast	54.2	55.6	247.7	175.8	375.2	76.2	15.9	17.8	8.9
Kenya	132.1	162.4	559.3	642.7	780.9	176.2	46.3	28.1	15.6
Madagascar	4283.8	> 10k	> 10k	> 10k	> 10k	6674.3	1662.0	223.1	152.2
Malawi	2547.3	5496.2	> 10k	> 10k	> 10k	3462.5	988.3	993.6	856.9
Mali	150.1	222.5	972.8	1361.4	1709.9	236.7	46.4	150.1	222.5
Mozambique	421.3	411.3	820.2	1194.4	948.2	484.0	214.2	299.8	251.0
Nepal	19.9	13.8	93.9	47.9	116.8	23.8	3.9	11.0	4.4
Niger	678.9	959.1	2939.1	4183.6	4257.3	936.1	257.9	678.9	959.1
Nigeria	517.8	1182.9	4976.7	> 10k	8603.3	748.6	155.6	275.3	241.7
Pakistan	109.5	112.7	590.7	542.4	830.0	154.3	22.8	14.4	5.2
Papua New Guinea	222.4	350.3	1266.8	2058.6	1953.5	318.5	80.6	149.1	142.2
Rwanda	280.9	335.9	899.7	593.0	893.9	279.2	113.8	94.5	62.9
Senegal	56.3	52.5	214.5	190.7	333.4	82.7	16.6	42.1	29.2
South Sudan	1009.5	864.8	5994.0	> 10k	NA	NA	193.0	526.8	290.9
Sudan	585.9	821.8	2406.0	> 10k	4500.9	1054.7	187.5	66.5	36.7
Tanzania	421.2	537.3	1480.9	1564.0	1809.7	497.8	163.0	122.4	80.3
Uganda	359.2	417.4	1029.6	1351.5	1234.0	431.9	143.1	134.1	91.7
Uzbekistan	313.3	482.6	1904.2	1326.4	1694.7	286.1	98.3	188.3	154.7
Yemen	82.5	18.0	363.7	2537.4	NA	NA	6.1	59.4	11.0
Zambia	414.3	513.9	1183.8	2283.8	1476.0	505.8	202.3	293.5	284.3
Zimbabwe	174.3	180.5	451.9	1156.5	736.9	266.1	77.5	129.9	110.7

Table A7: Income floor (in \$/day) financed by a 10% tax above \$10/day for major lower-income countries in 2030.

Growth scenario over 2022–2030	3%	3%	Projection		7%	7%	7% since 2015
HFCE rescaling		✓		✓		✓	
World	8.6	10.3	8.4	10.1	12.1	14.4	16.7
Low-Income Countries	1.9	2.2	1.5	1.7	3.3	3.6	4.4
Sub-Saharan Africa	2.1	2.7	1.9	2.5	3.1	3.9	5.3
Angola	2.1	3.2	1.7	2.8	3.1	4.6	7.8
Bangladesh	4.4	6.2	4.6	6.4	6.4	8.7	7.3
Benin	3.2	3.8	2.8	3.4	4.8	5.5	6.8
Burkina Faso	2.8	2.8	2.6	2.6	4.1	4.1	6.0
Burundi	0.9	0.9	0.7	0.7	1.3	1.3	2.8
Cameroon	3.1	3.5	2.6	3.0	4.5	5.1	7.7
Chad	1.8	1.8	1.6	1.6	2.8	2.8	7.1
D.R. Congo	1.0	1.0	0.9	0.9	1.6	1.6	2.7
Ethiopia	3.1	3.1	3.4	3.4	4.7	4.7	5.8
Ghana	3.1	5.2	2.9	5.0	4.6	7.3	6.7
Guinea	3.0	3.4	2.7	3.1	4.7	5.1	6.1
Haiti	1.3	2.3	1.0	2.0	1.9	3.3	4.3
India	3.8	5.9	3.9	6.1	5.6	8.2	7.1
Ivory Coast	4.1	5.6	3.6	5.1	6.0	7.9	8.1
Kenya	2.8	4.5	2.5	4.1	4.1	6.3	6.1
Madagascar	0.5	1.3	0.4	1.2	0.9	1.8	1.7
Malawi	0.9	1.1	0.8	1.0	1.3	1.6	2.4
Mali	3.0	3.0	2.5	2.5	4.4	4.4	7.6
Mozambique	1.3	1.4	1.2	1.3	1.8	2.0	3.3
Nepal	5.8	6.7	5.5	6.4	8.4	9.5	11.5
Niger	1.7	1.7	1.5	1.5	2.5	2.5	3.9
Nigeria	1.8	2.2	1.6	1.9	2.8	3.2	6.0
Pakistan	3.9	6.1	3.6	5.7	5.8	8.5	8.3
Papua New Guinea	2.0	2.3	1.7	2.0	3.0	3.4	5.8
Rwanda	2.0	2.7	2.0	2.7	2.9	3.8	3.9
Senegal	4.1	4.4	3.6	3.9	6.0	6.3	8.6
South Sudan	0.9	1.2	NA	NA	2.0	2.3	2.2
Sudan	1.9	3.3	1.5	2.7	2.8	4.5	7.6
Tanzania	1.9	2.6	1.8	2.4	2.8	3.6	4.2
Uganda	1.8	2.4	1.7	2.2	2.7	3.4	4.4
Uzbekistan	2.2	2.6	2.3	2.7	3.4	3.8	4.7
Yemen	3.6	4.0	NA	NA	7.8	8.3	7.2
Zambia	1.0	1.2	0.9	1.0	1.5	1.7	2.8
Zimbabwe	2.1	2.3	1.7	1.9	3.1	3.3	5.9

Table A8: Net gain per country of a global antipoverty tax above \$100/day, for most populous countries in 2030 after 3% growth since 2022. *Note that revenues are likely underestimated without HFCE rescaling, and the cost overestimated in lower-income countries with HFCE rescaling (as the extra income is wrongly attributed to the sole top 1%, which often concentrates the entire taxable base).*

Tax rate	1.2%	1.2%	0.3%	0.3%	10%	10%
HFCE rescaling		✓		✓		✓
Revenues (% global income)	0.15	0.41	0.04	0.10	1.20	3.42
International transfers	0.14	0.31	0.04	0.09	1.17	2.32
Income floor (\$/day)	2.17	3.31	1.44	2.14	4.29	7.00
Gini	0.62	0.71	0.62	0.71	0.60	0.65
Low-Income Countries	4.05	10.47	1.17	3.28	22.24	50.86
Sub-Saharan Africa	3.26	8.63	0.86	2.52	19.94	44.26
Algeria	0.00	-0.27	0.00	-0.07	0.04	-0.41
Angola	3.84	5.87	1.19	2.03	19.27	24.99
Argentina	-0.05	-0.37	-0.01	-0.09	-0.40	-3.07
Bangladesh	0.03	-0.22	0.00	-0.12	2.47	4.62
Brazil	0.00	-0.04	0.00	0.00	-0.26	-0.96
Canada	-0.20	-0.29	-0.05	-0.07	-1.65	-2.42
China	-0.01	-0.17	0.00	-0.04	-0.09	-0.97
Colombia	0.15	-0.30	0.08	-0.04	0.50	-2.96
D.R. Congo	11.53	32.00	3.45	11.05	54.50	126.84
Egypt	0.00	-0.81	0.00	-0.21	0.78	-4.83
Ethiopia	0.52	2.92	0.05	0.48	7.27	30.50
France	-0.16	-0.20	-0.04	-0.05	-1.36	-1.69
Germany	-0.19	-0.28	-0.05	-0.07	-1.60	-2.36
India	0.13	0.02	0.00	-0.10	4.62	6.07
Indonesia	0.00	-0.49	0.00	-0.16	1.42	-0.82
Iraq	0.00	-0.30	0.00	-0.07	0.07	-1.52
Italy	-0.15	-0.25	-0.04	-0.06	-1.30	-2.25
Japan	-0.14	-0.25	-0.04	-0.07	-1.15	-2.12
Kenya	1.05	1.92	0.19	0.36	11.05	14.07
Mexico	0.01	-0.57	0.00	-0.14	0.12	-4.45
Mozambique	12.59	28.74	4.08	10.77	52.87	103.65
Myanmar	0.03	0.35	0.00	0.01	1.93	12.30
Nigeria	2.51	9.51	0.39	2.10	22.09	59.44
Pakistan	0.01	-0.35	0.00	-0.16	3.38	4.76
Philippines	0.05	-0.36	0.00	-0.13	1.70	-0.24
South Korea	-0.10	-0.16	-0.02	-0.04	-0.82	-1.30
Spain	-0.12	-0.16	-0.03	-0.04	-1.03	-1.48
Sudan	2.23	5.17	0.33	0.95	23.73	33.87
Tanzania	3.27	9.65	0.45	2.08	28.28	52.23
Thailand	-0.01	-0.37	0.00	-0.09	-0.10	-2.88
Turkey	-0.0638	-0.50	-0.02	-0.13	-0.51	-4.04
Uganda	3.77	10.36	0.75	2.62	27.30	54.13
UK	-0.18	-0.30	-0.04	-0.07	-1.48	-2.46
USA	-0.42	-0.58	-0.11	-0.14	-3.51	-4.78
Vietnam	-0.01	0.00	0.00	-0.01	-0.01	0.59